

$$m_n/m_p = 1 + 1/720$$

# A Simple Approximation to the Neutron-Proton Mass Ratio

Relative difference: 0.00105%

Boris V. Kochenov

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## Abstract

The neutron-proton mass ratio is approximated by the simple expression

$$\frac{m_n}{m_p} \approx 1 + \frac{1}{720}.$$

This yields  $1 + 1/720 = 1.001388888\dots$  to be compared with the CODATA value  $m_n/m_p \approx 1.00137841946\dots$ . The relative difference is approximately 0.00105%.

The same integer term  $1/720$  also appears in other geometric relations within the Kochenov framework, including the refined form of the gravitational constant and the golden-ratio identity

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \varepsilon, \quad |\varepsilon| < 10^{-8}.$$

Since  $720 = 6!$ , the term is naturally associated with a compact factorial scale. Its recurrence across multiple independent relations suggests a possible discrete pattern, although the physical origin of this numerical agreement remains an open question.

## 1 Main Result

The neutron-proton mass ratio is given by the simple expression

$$\frac{m_n}{m_p} \approx 1 + \frac{1}{720}.$$

Numerically:

$$1 + \frac{1}{720} = 1.00138888888\dots$$

The CODATA value is

$$\frac{m_n}{m_p} \approx 1.00137841946\dots$$

The relative difference is approximately 0.00105%.

## 2 Context

The same term  $1/720 = 1/6!$  appears in other geometric relations within the Kochenov framework:

- The refined form of the gravitational constant  $G_{\text{Koch}}$
- The golden-ratio identity  $\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \varepsilon, |\varepsilon| < 10^{-8}$
- The fine-structure constant expression  $\alpha^{-1} = 144 - 7 - 1/720$

The recurrence of the factorial scale 720 across different domains (pure mathematics, gravity, nuclear physics) suggests a possible discrete geometric pattern, although its physical origin remains an open question.

## 3 Conclusion

A remarkably compact integer-based approximation comes very close to one of the fundamental mass ratios in physics. While the agreement is numerical rather than derived from a standard physical model, the repeated appearance of the same  $1/720$  term makes the pattern noteworthy and worth further examination.

## Related Publications

- [Platonic G: Gravitation from Geometry](#)
- [G\\_from\\_two\\_integers](#)
- [The Integer 31 as a Discrete Geometric Node in Kochenov G](#)
- [Geometric Origin of  \$10^{11}\$  in G \(Corrected\)](#)
- [A High-Precision Identity Linking  \$\phi^{12}/\sqrt{5}\$ , Fibonacci Numbers, and a Micro-Deviation of  \$1/720\$](#)

### Suggested citation:

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**Contact:** kochenovforwork@gmail.com